

Academic term	2026/27
Year of study	Year 3
Semester	1
Time / location	You have 95 minutes in total = 90 minutes for the test + 5 minutes to ZIP and upload

COMP H2037: Front End Development in-class lab test - SAMPLE

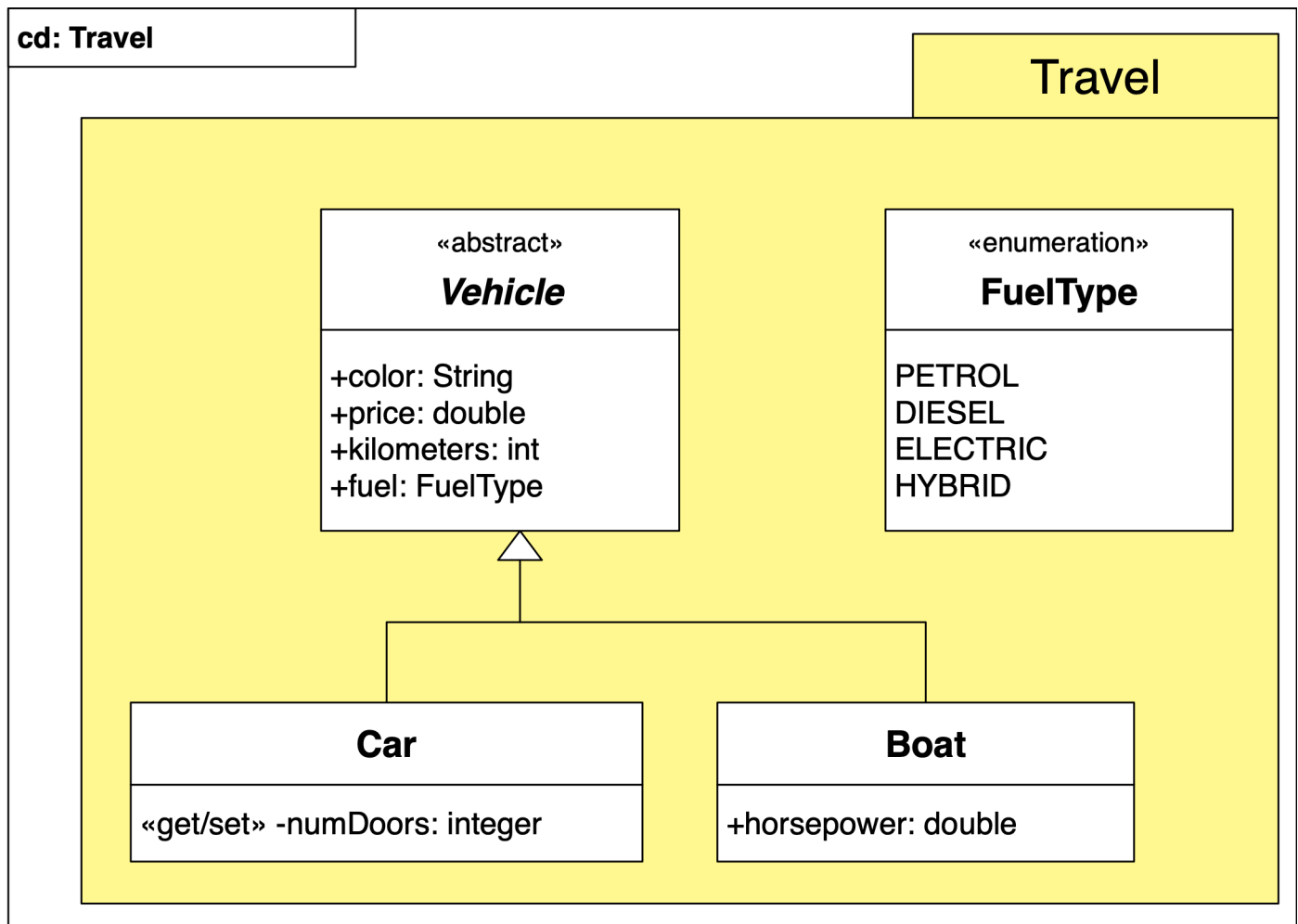
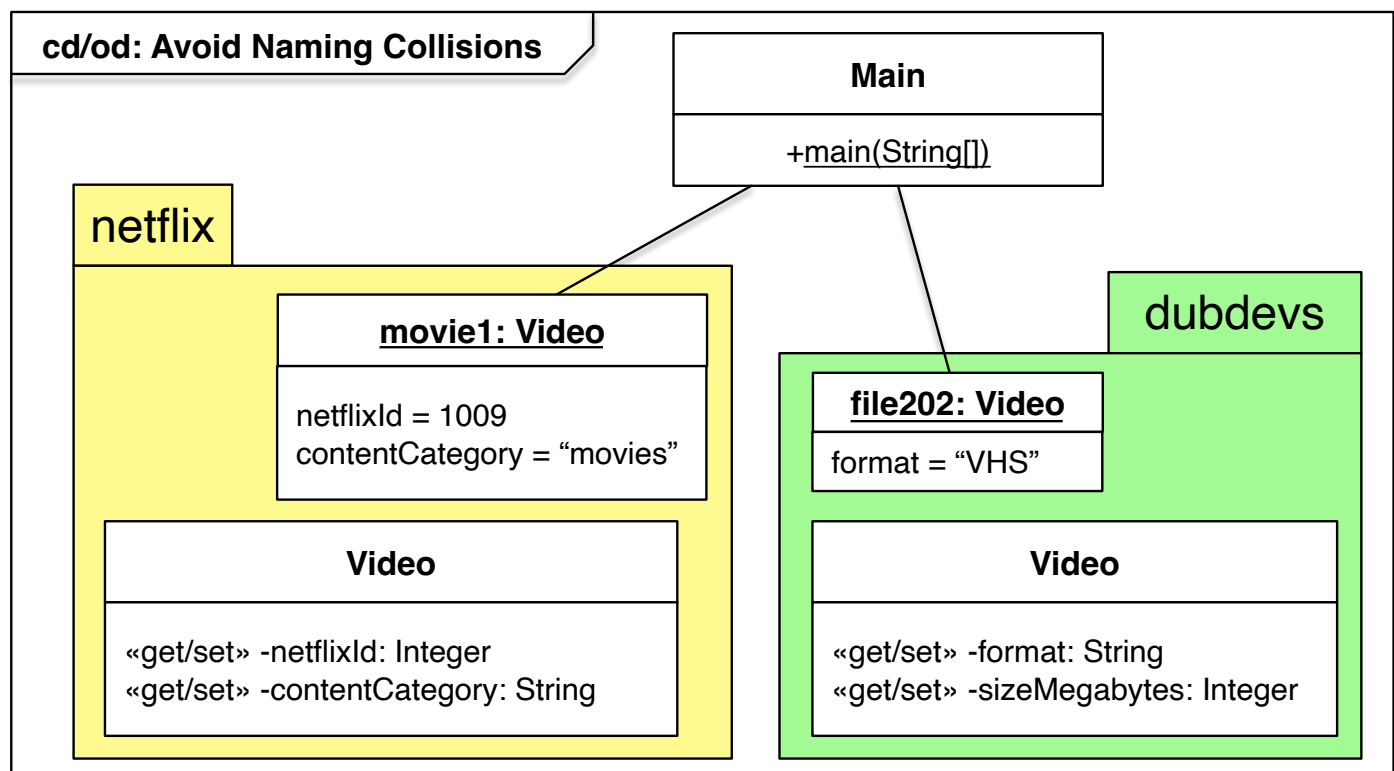
Internal Examiner(s)	Dr. Matt Smith
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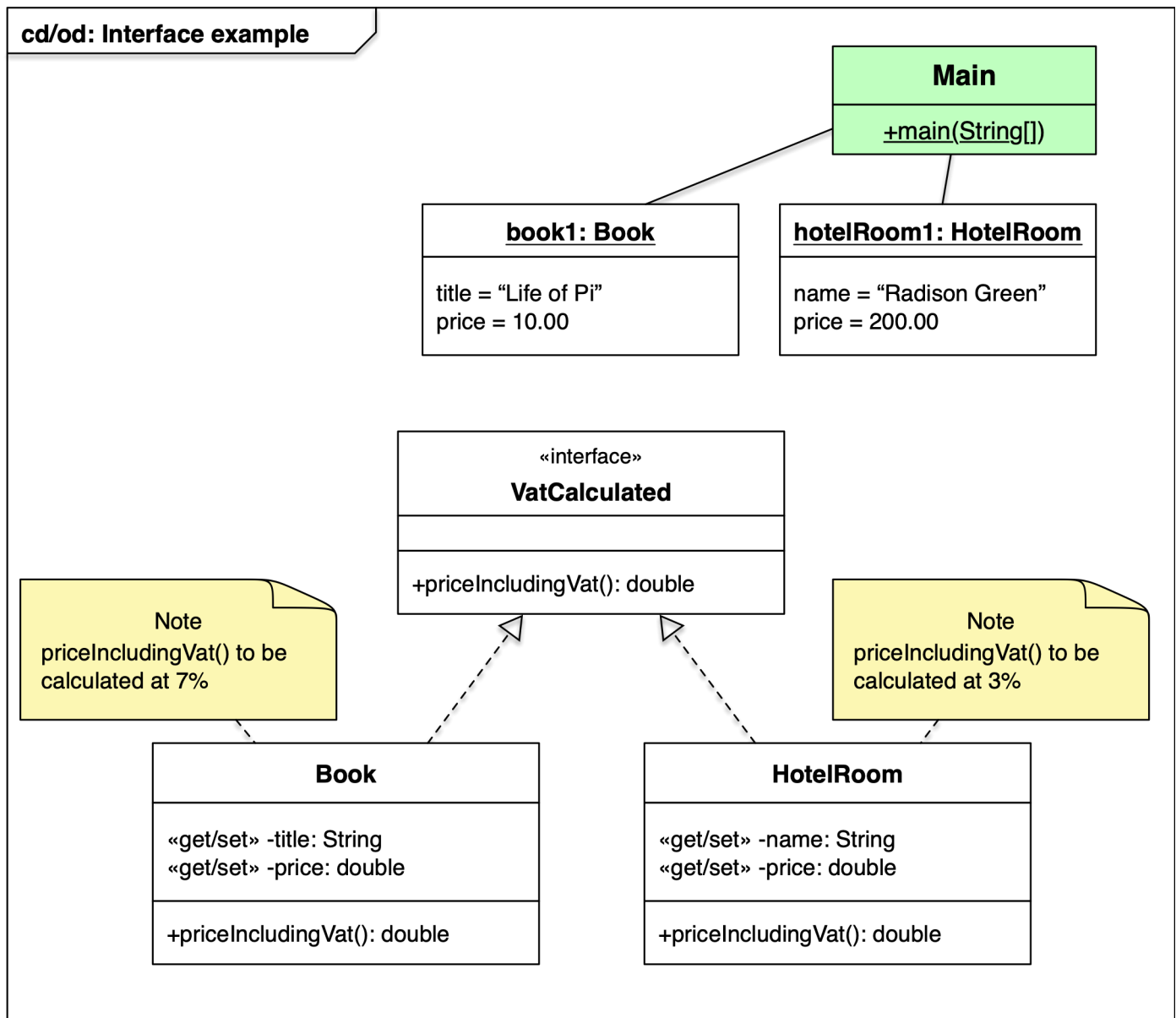
Instructions to candidates:

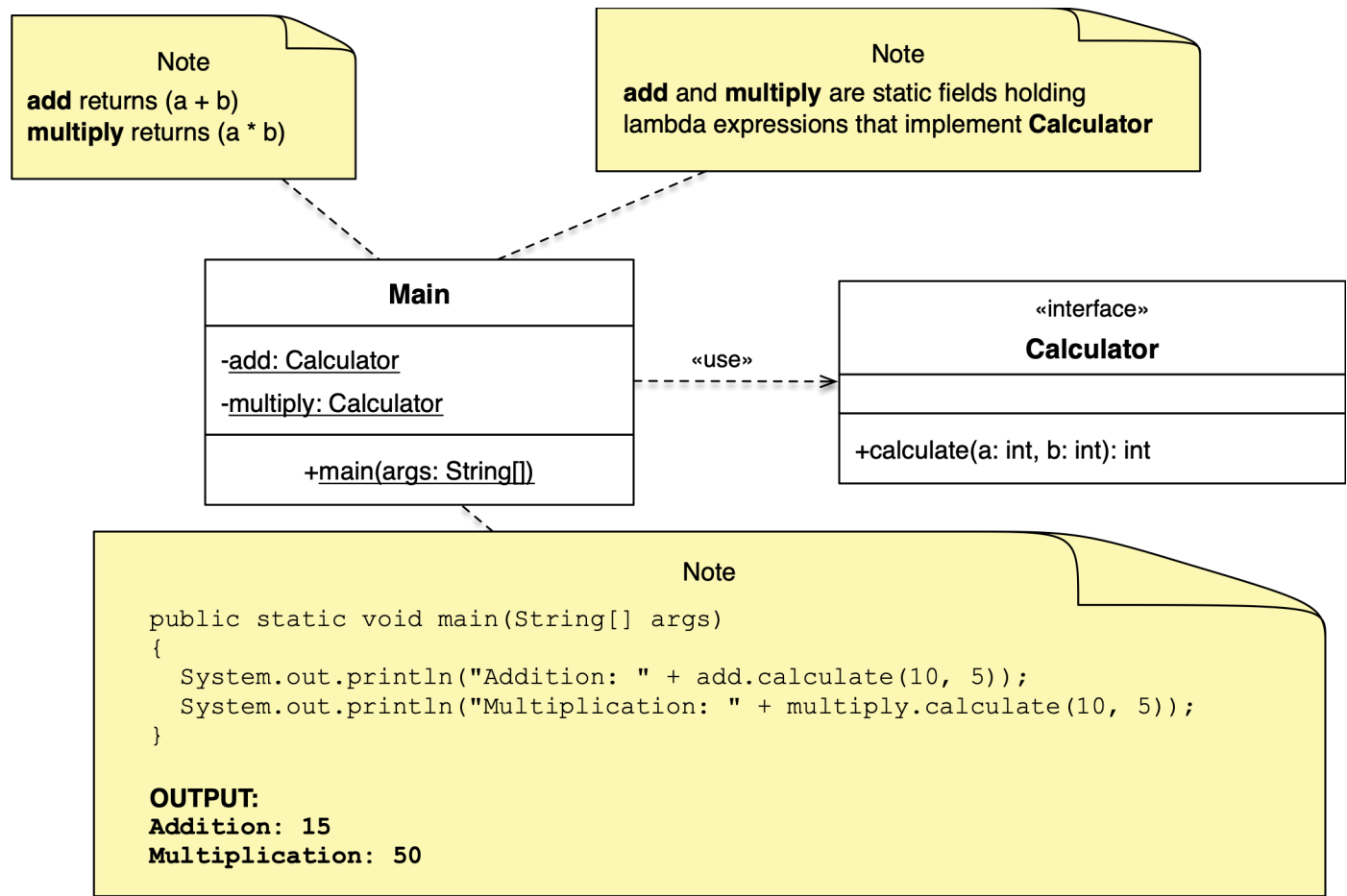
1.	Write Java classes to implement the diagram for each question. For some questions you may need to implement a main class that creates objects with property values as shown in the diagram.
2.	Create a folder for each question, and write your Java classes inside that folder. Question 1 code in folder 'question1,' question 2 code in folder 'question2', and so on...
3.	This is an <u>individual</u> test – all work should be your own. During the test you should NOT communicate with anyone apart from the lecturer (excepting medical / emergency circumstances).
4.	NO AI is permitted during this test
5.	NO communication with others is permitted (e.g. to NOT use phones, email etc.)
6.	You MUST use a university PC or your own laptop for this test. <i>NOTE: days/hours before the test ensure you know your password and can login!</i>
7.	This is an open-book test for paper-based & electronic materials. - You may look up paper and online examples/code reference during the test, as long as no AI help is used
8.	You may answer questions in any sequence.
9.	Answer ALL questions. The test is marked out of 100.

UPLOAD TO BRIGHTSPACE one ZIP folder:

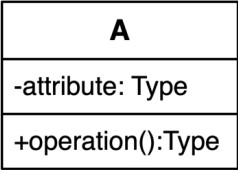
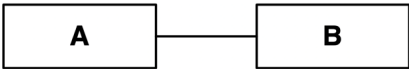
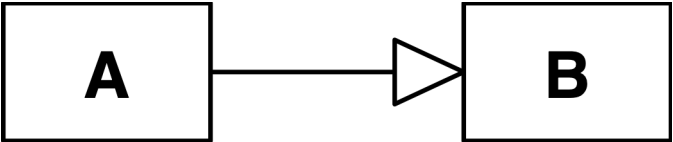
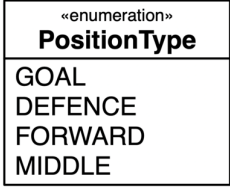
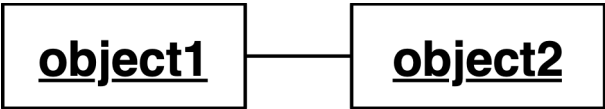
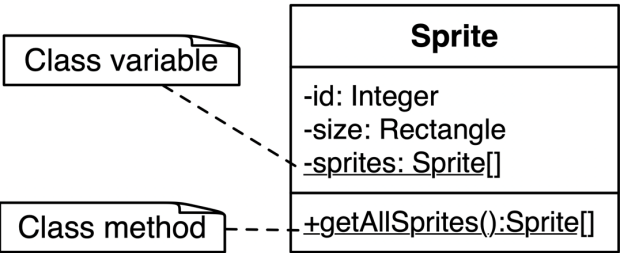
- **labtest.zip** – a ZIP folder containing all your question folders

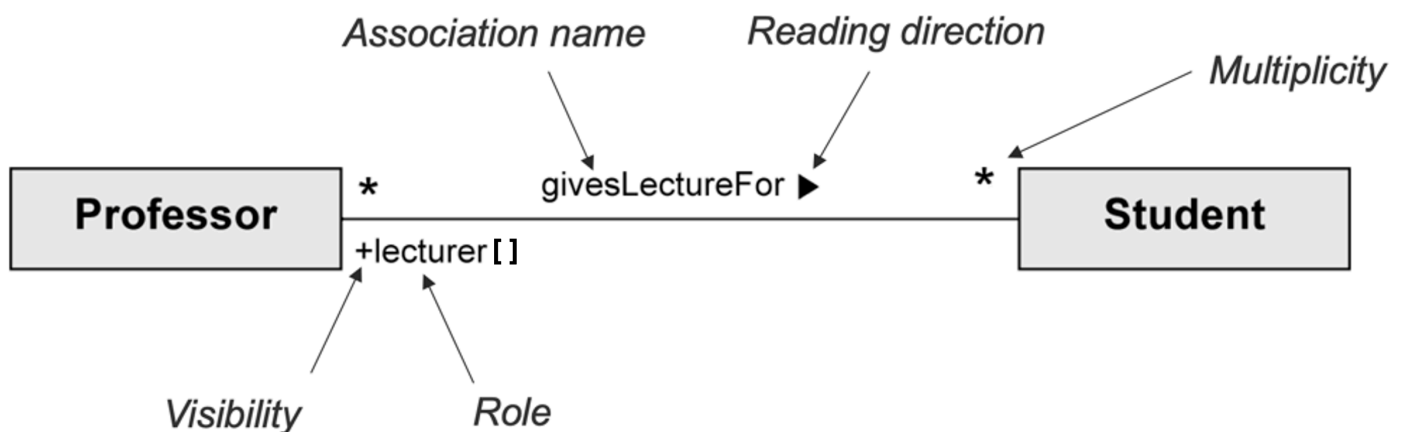
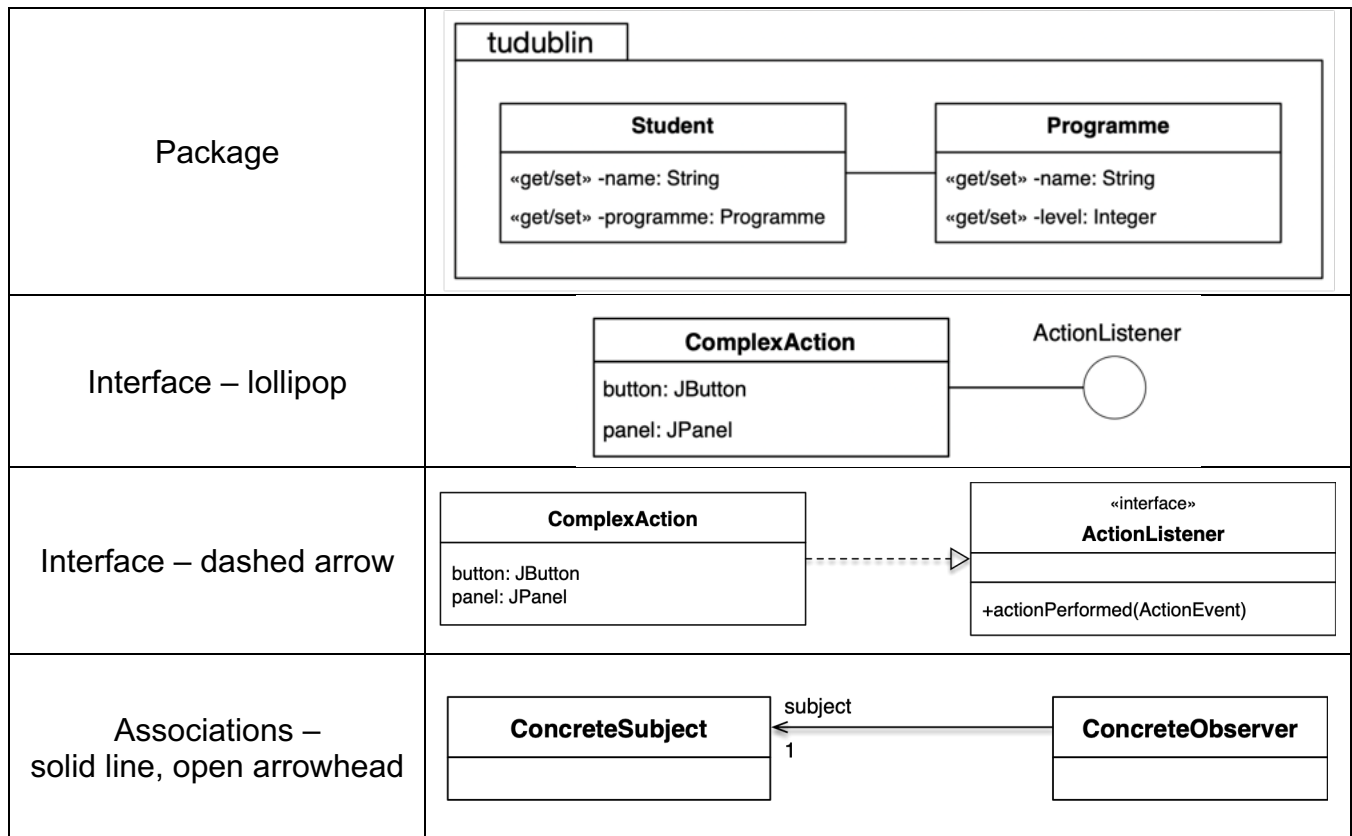
Question 1**20 marks****Question 2****20 marks**

Question 3**30 marks**

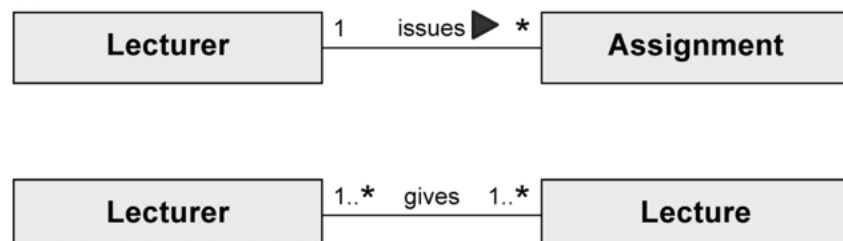
Question 4**30 marks**

APPENDIX: Summary of Class-Object-Package Diagram notation

Class Structure and behavior of set of objects	 <pre> classDiagram class A { -attribute: Type +operation():Type } </pre>
Class that cannot be instantiated - Both annotations may be found in diagrams	 <pre> classDiagram class A["{abstract} Class"] class B["«abstract» Class"] </pre>
Association between classes	 <pre> classDiagram A --- B </pre>
Generalization (inheritance) relationship A inherits from B (superclass may be abstract)	 <pre> classDiagram A -- > B </pre>
Enumeration	 <pre> classDiagram class PositionType["«enumeration» PositionType"] { GOAL DEFENCE FORWARD MIDDLE } </pre>
Object: instance of a class	 <pre> classDiagram class apple1_iphone["apple1:iphone"] { phoneNumber = "085-222-333" appleId = "jobs@me.com" network = net1 } </pre>
Association between objects	 <pre> classDiagram object1 --- object2 </pre>
Class (static) variable/method	 <pre> classDiagram class Sprite { -id: Integer -size: Rectangle -sprites: Sprite[] +getAllSprites():Sprite[] } Sprite --> "Class variable" Sprite --> "Class method" </pre>



- **Multiplicity:** Number of objects that may be associated with exactly one object of the opposite side



- Values for multiplicity:

- Min..Max
 - 0
 - 1
 - <n>
 - 0..*
 - *
- Means: 0..many
same as 0..*, i.e. 0..many